



Qualification Pack

CNC Milling Programmer

QP Code: CUTM/CSC/Q0202

Version: 1.0

NSQF Level: 5

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CUTM/CSC/Q0202: CNC Milling Programmer

Brief Job Description

CNC Milling Programmer is responsible for understanding the design requirements, material, generate CAM design for manufacturing of various parts using CNC lathe, inspect the parts to finalize and deliver the finished product. Also maintain good condition of the machine with regular maintenance

Personal Attributes

Personal attributes for a CNC Milling Programmer include strong attention to detail for interpreting designs and process planning, technical proficiency in CAM programming and machine setup, troubleshooting skills for preventive maintenance, and the ability to operate CNC machines efficiently. Additionally, a keen eye for quality control, inspection, and finalizing products ensures precision in delivering high-quality, finished components.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/CSC/N0205: Interpretation of designs, drawings and process planning](#)
2. [CUTM/CSC/N0206: CAM Programming, setting equipment Machining & Preventive maintenance of machines](#)
3. [CUTM/CSC/N0207: Operate CNC Milling machine and complete the machining process](#)
4. [CUTM/CSC/N0208: Inspect, finalize and deliver the product](#)
5. [CUTM/VSQ/N7805: Ensure and Implement Safety standards at Work Place](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Capital Goods
Sub-Sector	Advanced Machining
Occupation	Machining
Country	India
NSQF Level	5

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Credits	19
Aligned to NCO/ISCO/ISIC Code	NSQF/NCrF
Minimum Educational Qualification & Experience	Completed 3 year UG degree (B.E. or B. Tech degree in Mechanical, Production, Mechatronics) with NA of experience OR Completed 2nd year diploma after 12th (Mechanical or production) with NA of experience OR Completed 3 year diploma after 10th (Mechanical or production) with 1.5 years of experience relevant experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience relevant experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	19 Years
Last Reviewed On	NA
Next Review Date	30/01/2027
NSQF Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-05-CG-01976-2024-V1-CUTM
NQR Version	1.0

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CUTM/CSC/N0205: Interpretation of designs, drawings and process planning

Description

This NOS covers the skills and knowledge required to read and interpret mechanical engineering drawings and translate them into an actionable machining plan. It includes understanding CAD models, manufacturing tolerances, materials, and sequencing required for CNC milling operations.

Scope

The scope covers the following :

- This NOS includes interpretation of technical drawings, selection of materials and tools, and detailed process planning for CNC milling operations. It helps ensure that parts are manufactured to specification, reducing errors and improving production efficiency.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Interpret technical drawings and specifications of the product.
- PC2.** Identify critical features of the product.
- PC3.** Analyze manufacturability with the available resources.
- PC4.** Identify and prepare for potential assembly issues if any.
- PC5.** Identify and prepare for any maintenance issues that may arise during production.
- PC6.** Specify stock to be kept for different operations.
- PC7.** Identify critical sequences (with a high possibility of damage or unsafe practice).
- PC8.** Plan the production of parts according to specifications.
- PC9.** The importance of planning to improve efficiency.
- PC10.** Procedures to manufacture molds with the available resources.
- PC11.** Machining operations and their sequences.
- PC12.** Method of clamping work pieces.
- PC13.** Choices of cutting tools and cutting parameters.
- PC14.** Machine and work piece setting.
- PC15.** Measuring tools and equipment.
- PC16.** Bench work and assembly techniques.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** engineering drawings and geometric tolerances.
- KU2.** symbols, surface finish, and dimensional standards used in blueprints.



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- KU3.** sectional views, part lists, and annotations.
- KU4.** Bill of Materials (BoM) and manufacturing documents.
- KU5.** the steps in process planning and sequencing.
- KU6.** reference datum, datum shift, and part orientation.
- KU7.** common machining operations required per design.
- KU8.** raw material types and specifications.
- KU9.** how to identify critical dimensions for machining.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Interpret 2D and 3D part drawings for machining.
- GS2.** Identify dimensions, tolerances, and machining features.
- GS3.** Plan the machining process based on drawing analysis.
- GS4.** Select tools and materials based on drawing requirements.
- GS5.** Create operation sequences and time estimates.
- GS6.** Cross-check dimensions and design compatibility.
- GS7.** Communicate drawing changes or issues with engineers.
- GS8.** Record process plans and maintain documentation.
- GS9.** Use CAD software for reviewing and referencing part models.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	40	-	30
PC1. Interpret technical drawings and specifications of the product.	2	3	-	2
PC2. Identify critical features of the product.	2	3	-	2
PC3. Analyze manufacturability with the available resources.	2	3	-	2
PC4. Identify and prepare for potential assembly issues if any.	2	3	-	2
PC5. Identify and prepare for any maintenance issues that may arise during production.	2	3	-	2
PC6. Specify stock to be kept for different operations.	3	3	-	2
PC7. Identify critical sequences (with a high possibility of damage or unsafe practice).	3	3	-	2
PC8. Plan the production of parts according to specifications.	3	3	-	2
PC9. The importance of planning to improve efficiency.	2	3	-	2
PC10. Procedures to manufacture molds with the available resources.	2	1	-	2
PC11. Machining operations and their sequences.	2	2	-	2
PC12. Method of clamping work pieces.	1	2	-	2
PC13. Choices of cutting tools and cutting parameters.	1	2	-	2
PC14. Machine and work piece setting.	1	2	-	2
PC15. Measuring tools and equipment.	1	2	-	1
PC16. Bench work and assembly techniques.	1	2	-	1
NOS Total	30	40	-	30



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0205
NOS Name	Interpretation of designs, drawings and process planning
Sector	Capital Goods
Sub-Sector	
Occupation	Machining
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

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CUTM/CSC/N0206: CAM Programming, setting equipment Machining & Preventive maintenance of machines

Description

This NOS covers the use of CAM software for programming CNC milling machines, setting up tools and workpieces, and conducting preventive maintenance. It ensures that machines are configured correctly for optimal performance and longevity.

Scope

The scope covers the following :

- Includes programming and simulation in CAM, transferring programs to machines, physical setup, tool preparation, and routine maintenance. It ensures efficient and safe operation of CNC milling machines in a production environment.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Select the best sequence for machining each specific workpiece.
- PC2.** Program manually and in CAM software.
- PC3.** Transfer programs to machines.
- PC4.** Set work pieces and tools.
- PC5.** Apply the principles and processes of Computer Aided Manufacturing (CAM)
- PC6.** Set up and use machine centre input data into CNC machine controllers (tool offset, work offset, etc.)
- PC7.** Create machining programs in software, and transfer to machine controllers.
- PC8.** Test the finished products and assess them for accuracy in accordance with the specified drawings.
- PC9.** Machine each part of the dies, taking account of each plastic product's requirements.
- PC10.** measure pieces of work accurately.
- PC11.** set offsets according to measured size.
- PC12.** achieve the required geometry and finish.
- PC13.** Fabricate all parts to commercial standards using: Bench grinders.
- PC14.** Do preventative maintenance to avoid issues arising.
- PC15.** Make repairs when problem arise.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** CAM software interfaces and post-processing.
- KU2.** how to define machining operations in CAM.

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- KU3.** G-code and M-code fundamentals.
- KU4.** machine setup and zero referencing.
- KU5.** tool types, tool holders, and offsets.
- KU6.** about coolant systems and cutting parameters.
- KU7.** maintenance schedules and lubrication systems.
- KU8.** safety protocols in machine operation.
- KU9.** spindle speed, feed rate, and depth of cut.
- KU10.** common faults and diagnostics.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Create toolpaths using CAM software for CNC milling.
- GS2.** Simulate and verify machining programs before execution.
- GS3.** Set up tools, fixtures, and materials on the machine.
- GS4.** Load and run programs on CNC milling equipment.
- GS5.** Apply correct machining parameters for efficiency and safety.
- GS6.** Perform basic preventive maintenance (cleaning, lubrication, calibration).
- GS7.** Identify machine issues and raise maintenance requests.
- GS8.** Record setup sheets and tool data.
- GS9.** Maintain shopfloor cleanliness and tool management.
- GS10.** Ensure machine readiness for production cycles.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Select the best sequence for machining each specific workpiece.	3	4	-	-
PC2. Program manually and in CAM software.	3	4	-	-
PC3. Transfer programs to machines.	3	4	-	-
PC4. Set work pieces and tools.	3	4	-	-
PC5. Apply the principles and processes of Computer Aided Manufacturing (CAM)	3	4	-	-
PC6. Set up and use machine centre input data into CNC machine controllers (tool offset, work offset, etc.)	2	4	-	-
PC7. Create machining programs in software, and transfer to machine controllers.	2	4	-	-
PC8. Test the finished products and assess them for accuracy in accordance with the specified drawings.	3	4	-	-
PC9. Machine each part of the dies, taking account of each plastic product's requirements.	3	4	-	-
PC10. measure pieces of work accurately.	3	4	-	-
PC11. set offsets according to measured size.	3	4	-	-
PC12. achieve the required geometry and finish.	2	4	-	-
PC13. Fabricate all parts to commercial standards using: Bench grinders.	3	4	-	-
PC14. Do preventative maintenance to avoid issues arising.	2	4	-	-
PC15. Make repairs when problem arise.	2	4	-	-
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0206
NOS Name	CAM Programming, setting equipment Machining & Preventive maintenance of machines
Sector	Capital Goods
Sub-Sector	
Occupation	Machining
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

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CUTM/CSC/N0207: Operate CNC Milling machine and complete the machining process

Description

This NOS focuses on operating CNC milling machines by executing programmed instructions, adjusting parameters when needed, and ensuring that the final product meets specifications. It includes real-time monitoring and problem resolution during machining.

Scope

The scope covers the following :

- Involves running CNC programs, tool and parameter adjustments, part inspection during production, and efficient handling of operational challenges. This ensures continuous production flow and quality maintenance on CNC milling machines.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Follow their selected process strategy.
- PC2.** Follow a given process- strategy (external CNC programme)
- PC3.** Upload generated CNC programmes to CNC lathes and perform test runs.
- PC4.** Mount and align selected tools, holding devices.
- PC5.** Identify and designate the different machining processes on a CNC milling machine.
- PC6.** Optimize the machining strategy.
- PC7.** Define and adjust the cutting parameters as a function of the operation sequence, material type, type of operation, and CNC machine tool.
- PC8.** Start the cutting process from the raw material Solid block.

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- PC9.**
- Perform the following machining operations:
 - a. Facing
 - b. Roughing and finishing
 - i. External contours
 - ii. Island milling
 - iii. Milling channels
 - iv. Pocket (figurative)
 - v. Pocket (circular and rectangular)
 - vi. Taper ribs
 - c. Thread milling
 - i. Internal
 - ii. External
 - d. Canned cycles
 - i. Through hole boring
 - ii. Blind hole boring
 - iii. Reaming
 - iv. Tapping
 - v. Drilling
 - e. 3D machining operations
 - i. Roughing
 - ii. Finishing
 - Tapping
 - Drilling
 - 3D machining operations
 - Roughing
 - Finish
- PC10.** Avoid vibration in the machining sequences.
- PC11.** Apply burr-removal techniques on work pieces.
- PC12.** Optimize machining strategies.
- PC13.** Quickly react if anything goes wrong (making emergency stops)
- PC14.** Obtain dimensions, geometries, surface roughness by interacting with CNC milling machine.
- PC15.** Make the final parts conform to blueprints.
- PC16.** Report health, safety, and environmental issues to the appropriate personnel.
- PC17.** Report equipment failure to the appropriate personnel.
- PC18.** Note all the production related information on the respective register, log, work sheet.
- PC19.** Update all production information on the MIS system.
- PC20.** Store the CNC program against the batch number for future reference.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** machine operation sequences.
- KU2.** how to read and execute G/M code commands.
- KU3.** Learn monitoring techniques during machining.
- KU4.** use of control panels and feed overrides.
- KU5.** work holding techniques and alignment.



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- KU6.** tool life, wear, and replacement criteria.
- KU7.** the impact of speed/feed adjustments.
- KU8.** machining defects and corrections.
- KU9.** safety and emergency protocols.
- KU10.** documentation of production output and downtime.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Run CNC milling programs and monitor progress.
- GS2.** Adjust parameters during machining when required.
- GS3.** Replace worn tools and update offsets.
- GS4.** Handle production disruptions efficiently.
- GS5.** Perform visual and dimensional checks during operations.
- GS6.** Record machining cycle times and production data.
- GS7.** Maintain production logs and rejection reports.
- GS8.** Ensure consistent quality throughout production runs.
- GS9.** Follow SOPs and safety procedures strictly.
- GS10.** Communicate issues to supervisors and maintenance staff promptly.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Follow their selected process strategy.	2	3	-	-
PC2. Follow a given process- strategy (external CNC programme)	2	3	-	-
PC3. Upload generated CNC programmes to CNC lathes and perform test runs.	2	3	-	-
PC4. Mount and align selected tools, holding devices.	2	3	-	-
PC5. Identify and designate the different machining processes on a CNC milling machine.	1	2	-	-
PC6. Optimize the machining strategy.	2	3	-	-
PC7. Define and adjust the cutting parameters as a function of the operation sequence, material type, type of operation, and CNC machine tool.	2	3	-	-
PC8. Start the cutting process from the raw material Solid block.	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. - Perform the following machining operations: a. Facing b. Roughing and finishing i. External contours ii. Island milling iii. Milling channels iv. Pocket (figurative) v. Pocket (circular and rectangular) vi. Taper ribs c. Thread milling i. Internal ii. External d. Canned cycles i. Through hole boring ii. Blind hole boring iii. Reaming iv. Tapping v. Drilling e. 3D machining operations i. Roughing ii. Finishing - Tapping - Drilling 3D machining operations - Roughing - Finish	4	6	-	-
PC10. Avoid vibration in the machining sequences.	2	3	-	-
PC11. Apply burr-removal techniques on work pieces.	2	3	-	-
PC12. Optimize machining strategies.	2	3	-	-
PC13. Quickly react if anything goes wrong (making emergency stops)	2	3	-	-
PC14. Obtain dimensions, geometries, surface roughness by interacting with CNC milling machine.	2	3	-	-
PC15. Make the final parts conform to blueprints.	2	3	-	-
PC16. Report health, safety, and environmental issues to the appropriate personnel.	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. Report equipment failure to the appropriate personnel.	2	3	-	-
PC18. Note all the production related information on the respective register, log, work sheet.	2	3	-	-
PC19. Update all production information on the MIS system.	2	3	-	-
PC20. Store the CNC program against the batch number for future reference.	2	2	-	-
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0207
NOS Name	Operate CNC Milling machine and complete the machining process
Sector	Capital Goods
Sub-Sector	
Occupation	Machining
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

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CUTM/CSC/N0208: Inspect, finalize and deliver the product

Description

This NOS ensures that the finished machined product is inspected for quality, finalized as per specifications, and delivered appropriately. It includes the use of measuring tools, documentation, and handling practices.

Scope

The scope covers the following :

- Includes final inspection of parts, quality documentation, defect handling, product finishing, packaging, and delivery coordination. It ensures that the product is customer-ready and meets all quality and dimensional requirements.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Select the best sequence for machining each specific work piece.
- PC2.** Program manually and in CAM software.
- PC3.** Transfer programs to machines.
- PC4.** Set work pieces and tool.
- PC5.** Apply the principles and processes of Computer Aided Manufacturing (CAM)
- PC6.** Set up and use machine centre input data into CNC machine controllers (tool offset, work offset, etc.)
- PC7.** Create machining programs in software, and transfer to machine controllers.
- PC8.** Test the finished products and assess them for accuracy in accordance with the specified drawings.
- PC9.** Machine each part of the dies, taking account of each plastic product's requirements.
- PC10.** measure pieces of work accurately.
- PC11.** set offsets according to measured size.
- PC12.** achieve the required geometry and finish.
- PC13.** Complete the documentation as per ISO requirements.
- PC14.** Do preventative maintenance to avoid issues arising.
- PC15.** Use Statistical Process Control to ensure Quality of the process.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** geometric dimensioning and tolerancing (GD&T).
- KU2.** inspection techniques: visual, dimensional, and surface finish.
- KU3.** tools like micrometers, vernier calipers, CMMs.



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- KU4.** allowable tolerances and fit classifications.
- KU5.** about surface roughness and profile specifications.
- KU6.** how to fill inspection reports.
- KU7.** packaging requirements and labeling.
- KU8.** basics of customer delivery specifications.
- KU9.** traceability and batch control.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Inspect machined parts against drawing specifications.
- GS2.** Use precision instruments for quality checks.
- GS3.** Record inspection results in standard formats.
- GS4.** Identify and classify defects accurately.
- GS5.** Coordinate with QA team for approvals.
- GS6.** Perform minor polishing or deburring if needed.
- GS7.** Ensure correct packaging of finished goods.
- GS8.** Prepare parts for dispatch or storage.
- GS9.** Handle and transport finished components safely.
- GS10.** Maintain final inspection and delivery records.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	70	-	-
PC1. Select the best sequence for machining each specific work piece.	2	5	-	-
PC2. Program manually and in CAM software.	2	5	-	-
PC3. Transfer programs to machines.	2	5	-	-
PC4. Set work pieces and tool.	2	5	-	-
PC5. Apply the principles and processes of Computer Aided Manufacturing (CAM)	2	5	-	-
PC6. Set up and use machine centre input data into CNC machine controllers (tool offset, work offset, etc.)	2	5	-	-
PC7. Create machining programs in software, and transfer to machine controllers.	2	5	-	-
PC8. Test the finished products and assess them for accuracy in accordance with the specified drawings.	2	5	-	-
PC9. Machine each part of the dies, taking account of each plastic product's requirements.	2	5	-	-
PC10. measure pieces of work accurately.	2	5	-	-
PC11. set offsets according to measured size.	2	4	-	-
PC12. achieve the required geometry and finish.	2	4	-	-
PC13. Complete the documentation as per ISO requirements.	2	4	-	-
PC14. Do preventative maintenance to avoid issues arising.	2	4	-	-
PC15. Use Statistical Process Control to ensure Quality of the process.	2	4	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0208
NOS Name	Inspect, finalize and deliver the product
Sector	Capital Goods
Sub-Sector	
Occupation	Machining
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

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CUTM/VSQ/N7805: Ensure and Implement Safety standards at Work Place

Description

This NOS focuses on developing the ability to operate CNC machining centers safely and efficiently while adhering to health and safety standards. It includes creating CAM programs, selecting tools and cutting parameters, maintaining machinery, communicating effectively with teams and clients, and continuously upgrading skills to ensure excellence in manufacturing processes.

Scope

The scope covers the following :

- The scope encompasses CNC machine operation, safety compliance, program development, tool selection, maintenance, and effective collaboration with team members and customers. It promotes high professional standards through ongoing skill enhancement, problem-solving, and customer engagement while maintaining health and safety protocols throughout the manufacturing lifecycle.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** adhere to health and safety regulations in the workplace
- PC2.** proactively promote best practice in health and safety in the working environment
- PC3.** work independently on CNC machining centres
- PC4.** create manual and CAM programs for various types of machining
- PC5.** select suitable cutting parameters
- PC6.** select and set the most appropriate tools for the planned work
- PC7.** maintain all tools to ensure that they are in the best condition
- PC8.** communicate and collaborate effectively with colleagues, team members, and other professionals
- PC9.** engage with customers effectively, always prioritizing their needs
- PC10.** explain complex technical details to non-specialists
- PC11.** proactively engage in continuous professional development to promote excellence in the work and maintain expertise in current industrial practice analyze the manufacturing feasibility
- PC12.** successfully apply mathematical principles to complex industrial scenarios
- PC13.** demonstrate high levels of critical thinking

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Workplace health and safety regulations, and how to adhere to them.
- KU2.** Best practices in promoting health and safety proactively in CNC operations.

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- KU3.** Fundamentals of CNC machining processes and the safe operation of CNC machines.
- KU4.** CAM software programming principles and manual programming techniques.
- KU5.** Selection of appropriate cutting parameters and tools for various materials and tasks.
- KU6.** Maintenance procedures for tools and machines to ensure optimal performance and safety.
- KU7.** Communication protocols and collaborative practices within a technical team.
- KU8.** Techniques for engaging with customers and understanding their technical needs.
- KU9.** Simplifying and explaining technical information to non-specialists.
- KU10.** Importance of continuous professional development and industrial best practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Independently operate CNC machines with precision and safety.
- GS2.** Develop CAM programs suited for diverse machining requirements.
- GS3.** Select and calibrate tools and cutting parameters for optimal machining.
- GS4.** Maintain tools and machines in excellent working condition.
- GS5.** Collaborate effectively with teammates and interdisciplinary professionals.
- GS6.** Provide clear, simple explanations of technical content to clients or stakeholders.
- GS7.** Prioritize customer requirements and ensure satisfaction in service delivery.
- GS8.** Apply mathematical reasoning to solve complex machining problems.
- GS9.** Analyze the feasibility of manufacturing processes and adapt accordingly.
- GS10.** Exhibit strong critical thinking and decision-making in industrial scenarios.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	40	-	30
PC1. adhere to health and safety regulations in the workplace	2	3	-	2
PC2. proactively promote best practice in health and safety in the working environment	2	3	-	2
PC3. work independently on CNC machining centres	2	3	-	2
PC4. create manual and CAM programs for various types of machining	2	3	-	2
PC5. select suitable cutting parameters	2	3	-	2
PC6. select and set the most appropriate tools for the planned work	2	3	-	2
PC7. maintain all tools to ensure that they are in the best condition	2	3	-	2
PC8. communicate and collaborate effectively with colleagues, team members, and other professionals	2	3	-	2
PC9. engage with customers effectively, always prioritizing their needs	2	2	-	2
PC10. explain complex technical details to non-specialists	2	2	-	3
PC11. proactively engage in continuous professional development to promote excellence in the work and maintain expertise in current industrial practice analyze the manufacturing feasibility	3	4	-	3
PC12. successfully apply mathematical principles to complex industrial scenarios	3	4	-	3
PC13. demonstrate high levels of critical thinking	4	4	-	3
NOS Total	30	40	-	30



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/VSQ/N7805
NOS Name	Ensure and Implement Safety standards at Work Place
Sector	Cross Sectoral
Sub-Sector	
Occupation	Health and Safety
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIDH or email
- Assessment agencies send the assessment confirmation to TP looping AB/AA to accept in SIDH
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- AB monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.



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3. Assessment Quality Assurance levels/Framework:

- Question bank is created by the Subject Matter Experts (SME) are verified by the AB
- Questions are mapped to the specified assessment criteria.
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location.
- Centre photographs with signboards and scheme specific branding
- Video Call verification/online proctoring will be done during the assessment from AB.

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Minimum Percentage Required

- 70% overall to pass.
- Minimum 70% in each NOS

7. Reassessment: Candidates failing the assessment will be given one opportunity for reassessment after additional training.

8. Assessor Details: ToA Certified Assessor with relevant experience.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 70

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a

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whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/CSC/N0205.Interpretation of designs, drawings and process planning	30	40	-	30	100	20
CUTM/CSC/N0206.CAM Programming, setting equipment Machining & Preventive maintenance of machines	40	60	-	-	100	20
CUTM/CSC/N0207.Operate CNC Milling machine and complete the machining process	40	60	-	-	100	20
CUTM/CSC/N0208.Inspect, finalize and deliver the product	30	70	-	-	100	15
CUTM/VSQ/N7805.Ensure and Implement Safety standards at Work Place	30	40	-	30	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	190	300	-	60	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.